

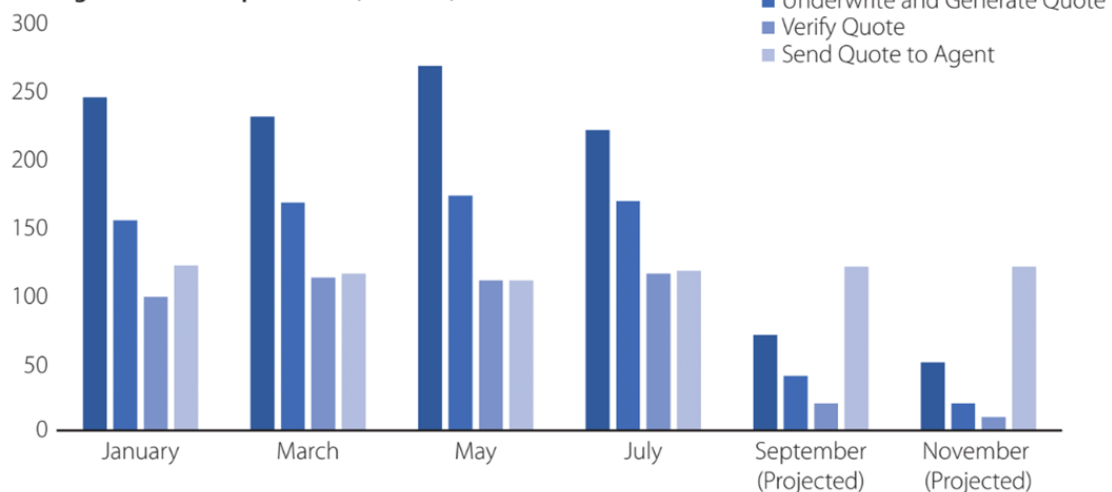
Workshop Activities

Activity 1 – Chan Life Insurance

Consider the clustered column chart for the Chan Life Insurance underwriters audience shown below.

New IT System Projected to Decrease Processing, Underwriting, and Verification Times

Average Time to Complete Task (minutes)



The chart displays the task times for responses to quotes for four historical data points (January, March, May, and July) and two projected ones (September and November). The data for the chart is shown below.

Task	Average Time to Complete Task (minutes)					
	January	March	May	July	September (Projected)	November (Projected)
Process Request for Quote	244	230	267	220	70	50
Underwrite and Generate Quote	154	167	172	168	40	20
Verify Quote	98	112	110	115	20	10
Send Quote to Agent	121	115	110	117	120	120
Total Response Time	617	624	659	620	250	200

However, the clustered column chart appears cluttered and unnecessarily increases the cognitive load by forcing the audience to navigate back and forth to understand the provided insight

Complete the following steps:

1. Recreate the clustered column chart using the data provided in the above table.
2. Create an alternative visualisation that addresses the cluttering and cognitive load issues.

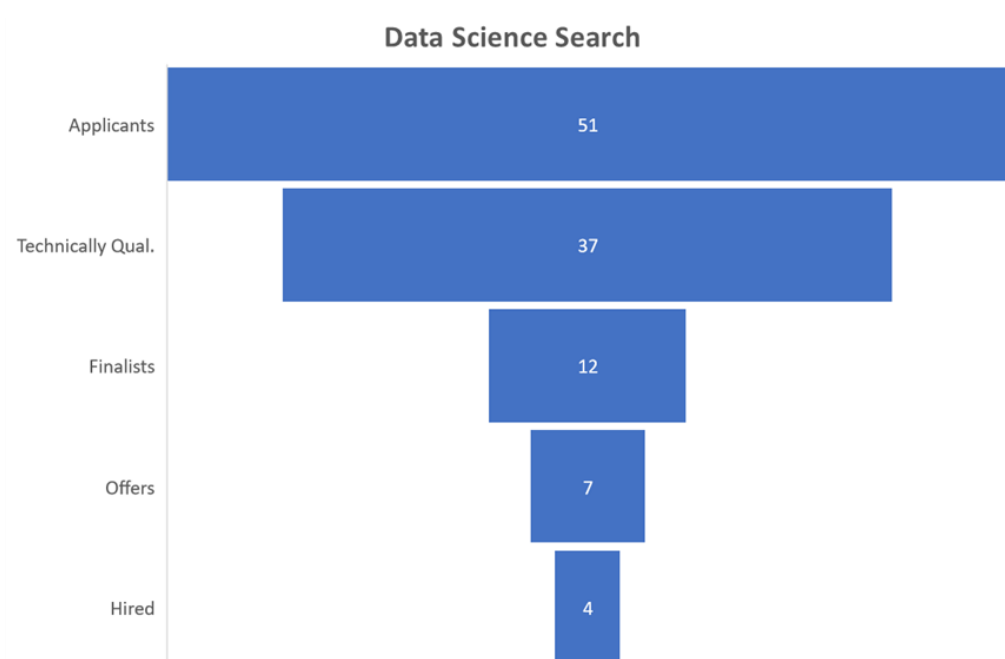
Activity 2 – Creating a Funnel Chart using Excel

This activity creates a funnel chart using data from the *DataScienceSearch* data file. The file contains data on a sample of applicants for a data science job.

Open the *DataScienceSearch* data file and complete the following steps:

- Step 1.
 - Select cells A1:B6.
- Step 2.
 - Click the Insert tab on the Ribbon
- Step 3.
 - Click the Insert Waterfall, Funnel, Stock, Surface or Radar Chart button in the Charts group
- Step 4.
 - When the list of subtypes appears, click the Funnel Chart button

Edit the chart to improve its appearance. The result should look similar to the following chart.



In what other situations would a funnel chart be useful?

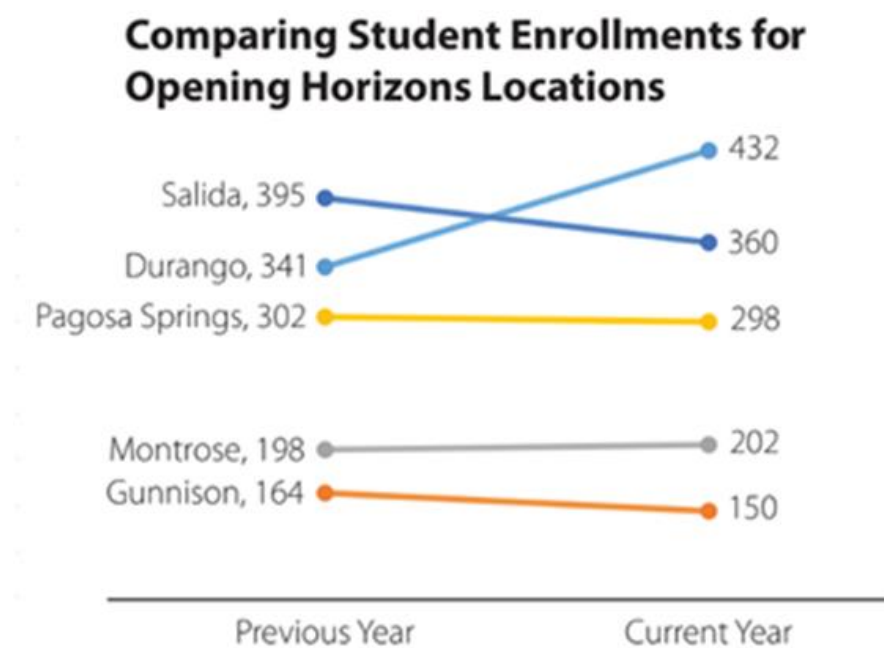
Activity 3 – Creating a Slope Chart using Excel

This activity creates a slope chart using data from the *OpeningHorizons* data file. The file contains data that compares student enrolments at 5 locations from the start of the current year to the start of the previous year.

Open the *OpeningHorizons* data file and complete the following steps:

- Step 1.
 - Select cells A2:C7
- Step 2.
 - Click the Insert tab on the Ribbon
 - Click the Insert Line or Area Chart button in the Charts group
 - Select Line with Markers
- Step 3.
 - When the default chart appears in Excel, right-click the chart and choose Select Data
 - When the Select Data Source dialog box appears, click the Switch Row/Column button
 - Click OK
- Step 4.
 - Click anywhere on the chart and click the Chart Elements button
 - Deselect the check box for Gridlines
 - Deselect the check box for Legend
 - Select the check box for Data Labels and choose Right
- Step 5.
 - Double click the “395” data label for Salida so that only this data label is selected
 - When the Format Data Label task pane appears, click the Label Options icon
 - Click Label Options, select the check box for Series Name under Label Contains and select Left under Label Position
- Step 6.
 - Repeat Step 5 for the “341,” “302,” “198,” and “164” data labels.
- Step 7.
 - Click the vertical axis values and click Delete
- Step 8.

- Click the chart title box and change the title to Comparing Student Enrolments for Opening Horizons Locations
- Click the Home tab on the Ribbon
- Click the Align Left button in the Alignment group to left justify the chart title
- Change the chart title font to Calibri 16 pt Bold
- Drag the chart title box to align with the text to the left of the chart
- Step 9.
 - Adjust the size of the chart area and the sizes of the text boxes for the data labels so that the finished chart appears similar to the following chart.



Describe the insights that can be drawn from this chart.

Activity 4 – Engineering Graduate Salaries at Empire State University

Empire State University (ESU) collects salary data for its engineering college graduates. ESU collects data on starting salaries for its graduates and then again five years post-graduation. The table below shows the median starting salaries and median salaries five years post-graduation by engineering major. Create a slope chart to display these data.

Engineering Major	Salaries	
	Starting	5 Years Post Graduation
Chemical	\$72,500	\$91,400
Civil	\$61,000	\$70,300
Computer	\$68,400	\$83,600
Electrical	\$64,800	\$87,400
Industrial	\$57,400	\$74,000
Mechanical	\$62,900	\$77,100

Describe the insights that can be drawn from the chart.

Activity 5 – Creating a Dot Matrix Chart using Excel

Unemployment data from the United States suggests that in the year 2020, due to the Covid 19 pandemic, 40 million jobs were lost. One way to visualize this is to use a dot matrix chart which can be created in Excel. Visualising 40 million lost jobs requires a 20 × 20 matrix of dots.

Open a blank Excel worksheet and complete the following steps:

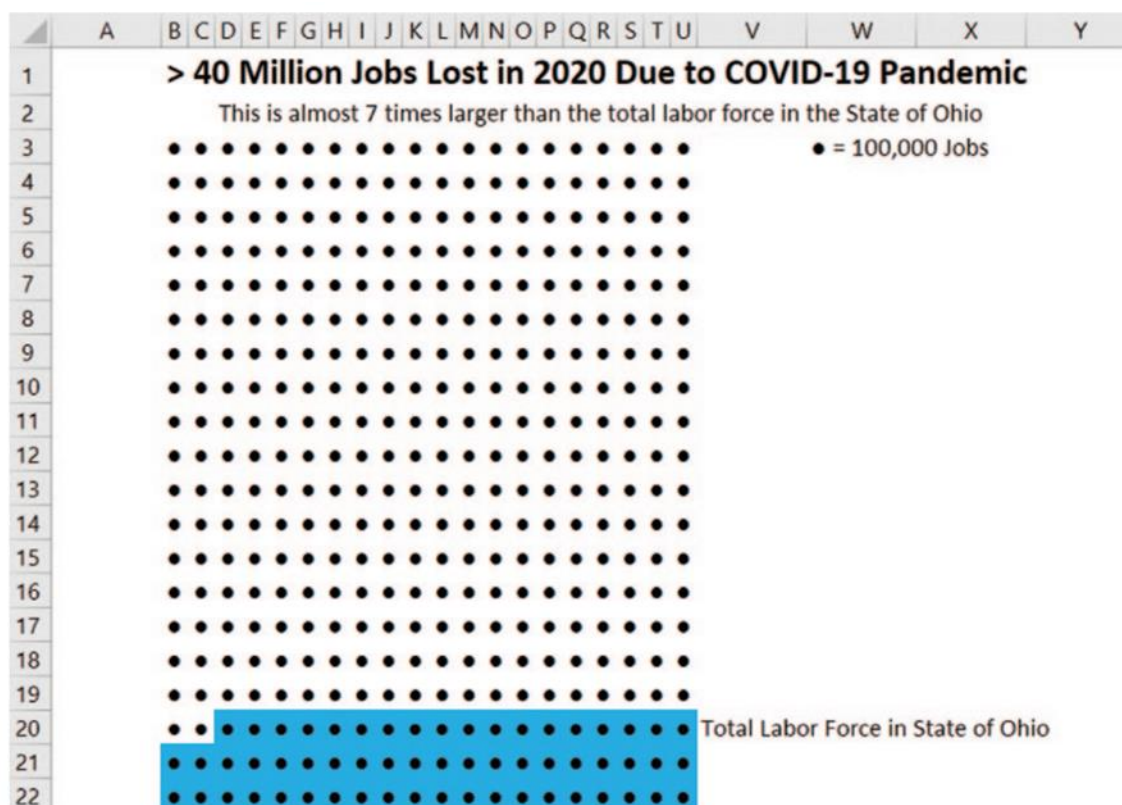
- Step 1.
 - Select cell B3 in a blank worksheet in Excel
 - Click the Insert tab on the Excel Ribbon
 - In the Symbols group, click Symbol
- Step 2.
 - When the Symbol dialog box opens:
 - Select Geometric Shapes in the Subset: box and click the filled dot
 - Click Insert to insert this dot in cell B3
 - Click Close to exit the Symbol dialog box
- Step 3.
 - Copy cell B3 to cells B3:U22
- Step 4.
 - Highlight Columns B:U and double-click the edge of one of the columns to reduce the spacing of the columns
- Step 5.
 - Click View in the Excel Ribbon and deselect the check box for Gridlines

Additional context for these data can be included by adding additional details to the chart using the following steps:

- Step 6.

- Change the fill colour of cells D20:U20 and B21:U22 by clicking the Home tab on the Ribbon and then changing the Fill Color to blue
- Step 7.
 - Select cell B1 and enter the text 'More than 40 Million Jobs Lost in 2020 Due to COVID-19 Pandemic'
 - Select cell B1, click Home on the Ribbon, and change the font to 16 point Bold
- Step 8.
 - Select cell D2 and enter the text 'This is almost 7 times larger than the total labour force in the State of Ohio' using 11 point font
- Step 9.
 - Select cell W2 and enter the text = 100,000 Jobs
- Step 10.
 - Select cell V20 and enter the text 'Total Labour Force in State of Ohio'

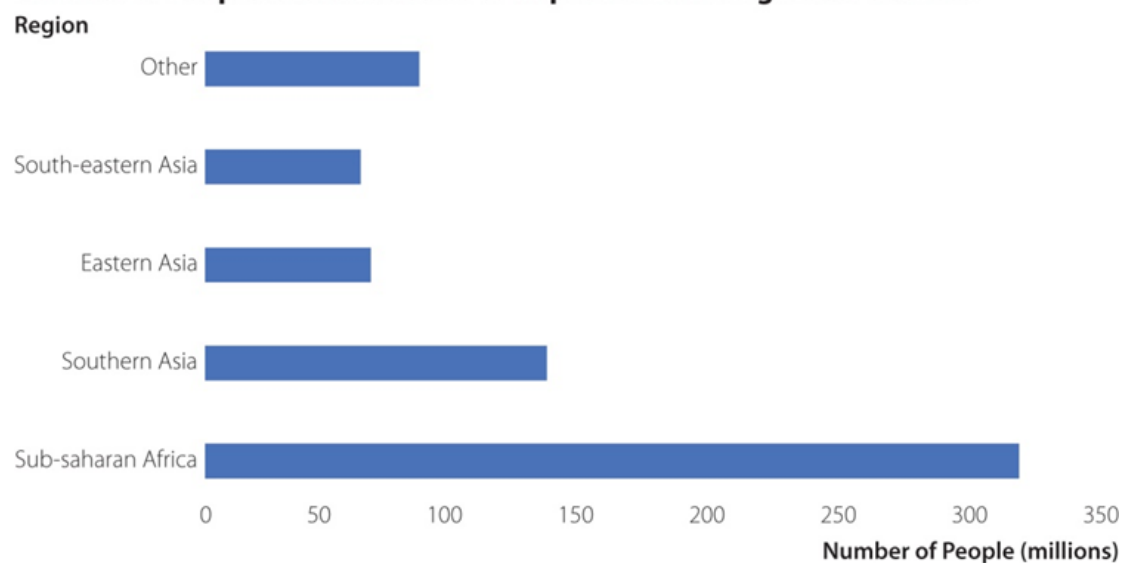
The result should look similar to the following chart.



Activity 6 – Lack of Access to Improved Drinking Water

Improved drinking water sources are defined as drinking water sources that are protected from outside contamination. According to the World Health Organization, 663 million people around the world lack access to sources of improved drinking water, which puts them at high risk for infection and illness. The bar chart below shows the number of people in each region that are estimated by the World Health Organization to lack access to improved drinking water sources.

Number of People Without Access to Improved Drinking Water Sources



- Add a BAN to the bar chart in the file *DrinkingWaterChart* so that the total number of people in the world who do not have access to improved drinking water sources is emphasized in this chart.
- To help create empathy with the data for the audience, create a dot matrix chart to represent the data in the file *DrinkingWaterChart*. Use ● = 1 million people for your dot matrix chart. (Hint: To represent 663 million people who lack access to improved drinking water resources, you will need a matrix that contains 663 ●. You can create this by using a 25 x 26 matrix and then adding an additional partial row of 13 ●.)
- Differentiate the colours of the ● based on the region of the world. Include a legend on your modified dot matrix chart.
- The population of the United States is approximately 330 million people. To help give the audience an even better sense of scale for the number of people around the world who lack access to improved drinking water, use the Gestalt principle of enclosure by using the Fill Color function in Excel to shade

the cells containing the number of ● that correspond to the population of the United States and label this on your chart.

Activity 7 – Creating a Stock Chart using Excel

This activity creates a stock chart using data from the *Verizon* data file. The file contains stock price data for a telecommunications company.

Open the *Verizon* data file and complete the following steps:

- Step 1.
 - Select cells A1:A6. Hold down the control key (Ctrl) and also select cells C1:E6
- Step 2.
 - Click the Insert tab on the Ribbon
- Step 3.
 - Click the Insert Waterfall, Funnel, Stock, Surface or Radar Chart button in the Charts group
 - When the list of subtypes appears, click the High-Low-Close button
 - Edit the chart to improve its appearance.

The following steps add the closing price labels and markers.

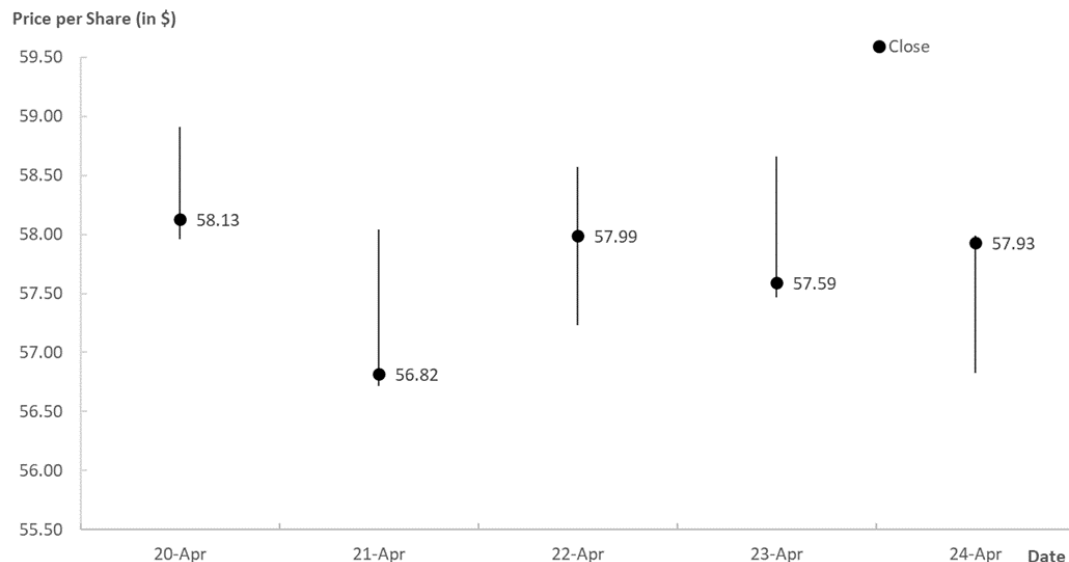
- Step 4.
 - Click the Chart Elements button and select Data Labels.

Step 4 places three sets of labels on each vertical bar (highest, closing and lowest price per share). The following steps clean up the display.

- Step 5.
 - Click any of the high price per share labels and press the Delete key. Do the same for the low price per share labels.
- Step 6.
 - On one of the vertical lines, click a data point directly next to one of the closing price labels.
- Step 7.
 - When the Format Data Series task pane appears, click the Fill &Line button, then click Marker.
 - Under Fill, select Solid fill and to the right of Color, select black from the drop-down menu.

- Under Border, select Solid line, and to the right of Color, select black from the drop-down menu.
- To the right of Width, select 3 pt.

The result should look similar to the following chart.



What insights does the chart provide?

Activity 8 – Creating a Waterfall Chart using Excel

This activity creates a waterfall chart using data from the *GossamerGP* data file. A waterfall chart shows the cumulative effect of positive and negative changes on a variable. The *GossamerGP* file contains data on the costs, revenue and gross profit of Gossamer Industries.

Open the *GossamerGP* data file and complete the following steps:

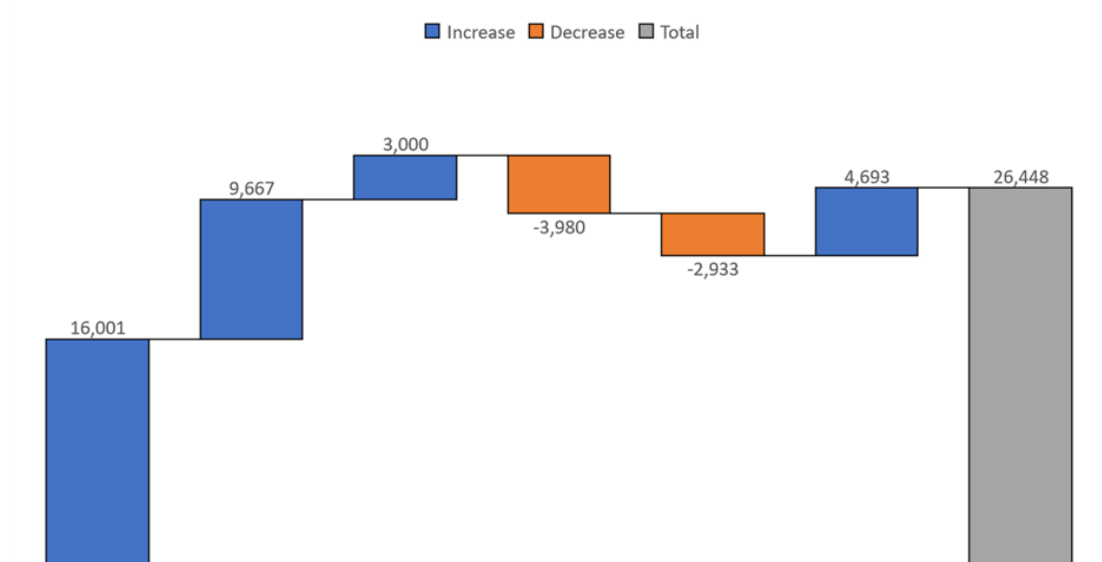
- Step 1.
 - Select cells A2:H2. Hold down the control key (Ctrl) and also select cells A5:H5
- Step 2.
 - Click the Insert tab on the Ribbon
- Step 3.
 - Click the Insert Waterfall, Funnel, Stock, Surface or Radar Chart button in the Charts group

- When the list of subtypes appears, click the Waterfall button

In the initial chart, notice that the Total has been treated like another month. Continue with the following steps:

- Step 4.
 - Double-click the column Total to activate the Format Data Series task pane, and then click the column of data again to activate the Format Data Point task pane
- Step 5.
 - When the Format Data Point task pane appears, click the Series Options button
 - Select the check box for Set as total

Edit the chart to improve its appearance. The result should look similar to the following chart.



Activity 9 – Air Transat Flight Delay

Consider the Air Transat Flight Delay case study:

In July 2017, Air Transat diverted passengers aboard two flights to Ottawa. While the airline had no control over the diversion itself, they were responsible for the fair treatment of their passengers following landing. Yet passengers reported being stranded on the tarmac for over four hours, on one flight without any food, air conditioning, or water—the incident caused at least one passenger to call 911 to force the airline to act.

Following the incident, the airline was fined and held responsible by the Canadian Transportation Agency (CTA.) During the hearing, customers and the press noted that

the airline and airport continually deflected responsibility and tried to blame each other further frustrating the passengers.

See the following article for case details:

<https://www.cbc.ca/news/canada/ottawa/air-transat-delay-decision-day-1.4426687>

Using Freytag's Pyramid approach to storytelling, develop a storyboard layout for the Air Transat case using the slide sorter view of PowerPoint.